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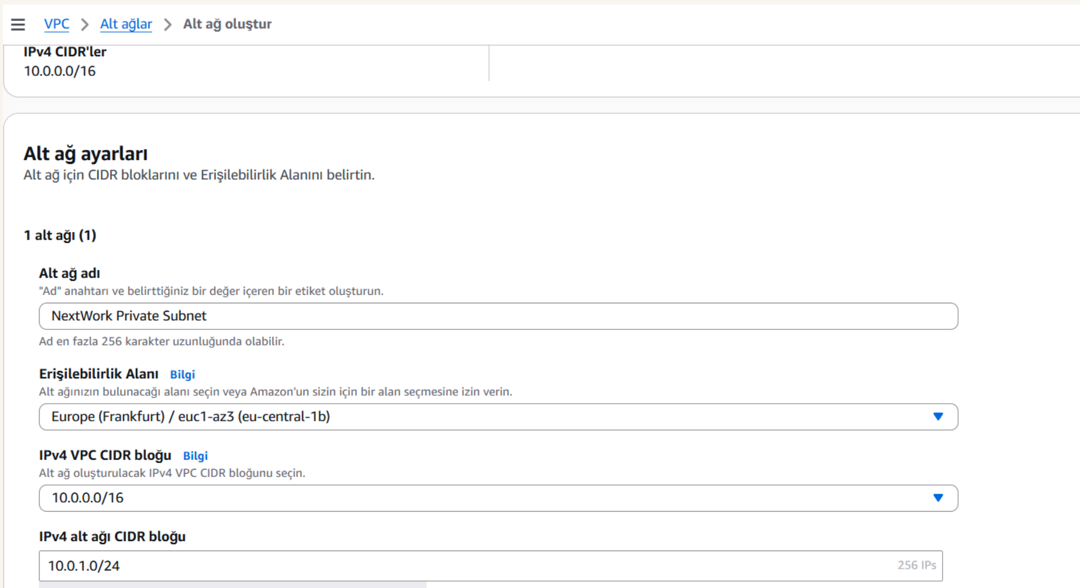
Creating a Private Subnet

E

Pelin Ece Burgun



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is a logically isolated network environment within AWS where you have full control over your IP address ranges, subnets, route tables, gateways, and security settings. It is useful because it lets you design a network architecture that mirrors an on-premises data center — but with the scalability and flexibility of the cloud. Without a VPC, all your AWS resources would sit in a shared, flat network with no isolation, which is a serious security risk in any real-world workload.

How I used Amazon VPC in this project

In this project, we used Amazon VPC to build a segmented, secure network from scratch. We created a VPC with both a public and a private subnet, attached an internet gateway to enable external access for the public subnet only, and configured separate route tables to enforce routing rules for each subnet. We also applied a custom Network ACL to the private subnet to restrict traffic strictly to internal VPC communication, ensuring the private subnet remained completely isolated from the internet.

One thing I didn't expect in this project was...

I didn't expect that every new subnet is automatically associated with the main route table by default. It made me realize how easy it is to accidentally expose a 'private' subnet to the internet if you don't explicitly create and assign a separate route table.

This project took me...

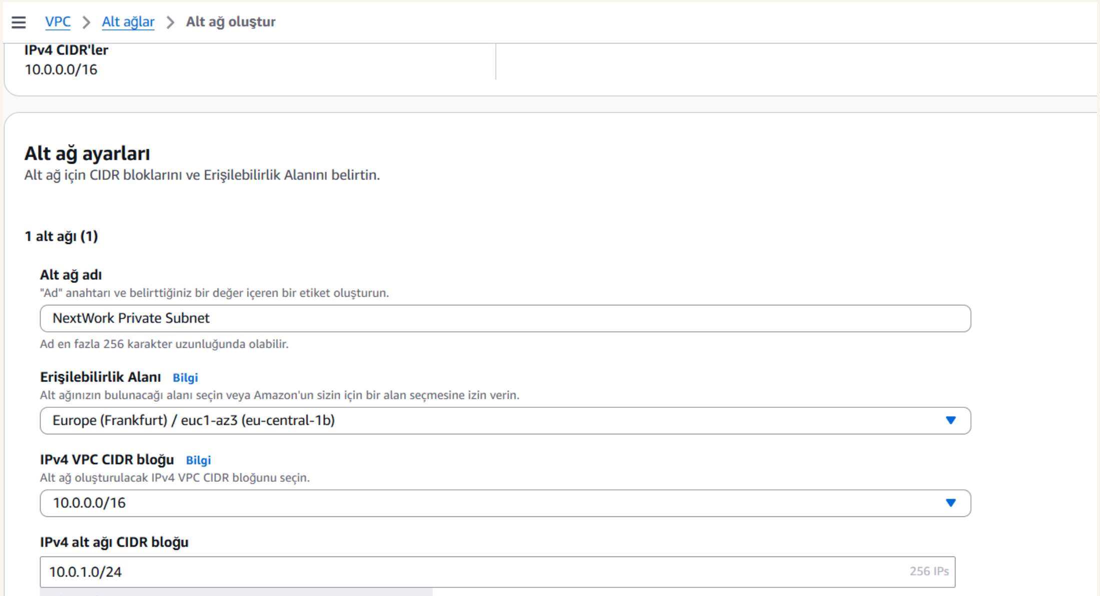
This project took me approximately 90 minutes.

Private vs Public Subnets

A public subnet has a route to an internet gateway, meaning resources inside it can send and receive traffic from the internet. A private subnet has no such route — resources inside it are isolated from the internet and can only communicate within the VPC or through controlled channels like a NAT gateway or VPN.

Private subnets exist to protect sensitive resources — such as databases, backend servers, or internal services — from direct internet exposure. Not everything in a network should be publicly reachable. By placing these resources in a private subnet, you significantly reduce the attack surface and enforce the principle of least privilege at the network level.

Private and public subnets cannot share the same route table if you want to maintain their separation. A public subnet's route table contains a route pointing to the internet gateway; a private subnet must have its own route table without that route. Sharing a route table would unintentionally expose the private subnet to the internet, defeating its entire purpose.

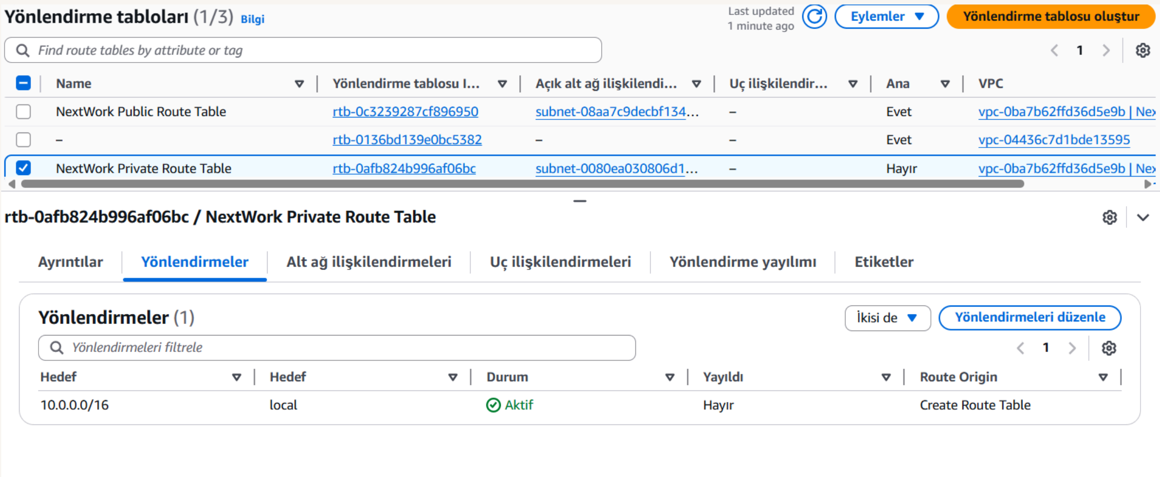

VPC > Alt ağlar > Alt ağ oluştur
IPv4 CIDR'ler
10.0.0.0/16
Alt ağ ayarları
Alt ağ için CIDR bloklarını ve Erişilebilirlik Alanını belirtin.
1 alt ağı (1)
Alt ağ adı
"Ad" anahtarı ve belirttiğiniz bir değer içeren bir etiket oluşturun.
NextWork Private Subnet
Ad en fazla 256 karakter uzunluğunda olabilir.
Erişilebilirlik Alanı Bilgi
Alt ağınızın bulunacağı alanı seçin veya Amazon'un sizin için bir alan seçmesine izin verin.
Europe (Frankfurt) / euc1-az3 (eu-central-1b)
IPv4 VPC CIDR bloğu Bilgi
Alt ağ oluşturulacak IPv4 VPC CIDR bloğunu seçin.
10.0.0.0/16
IPv4 alt ağı CIDR bloğu
10.0.1.0/24
256 IPs

A dedicated route table

By default, any new subnet in a VPC is automatically associated with the main route table of that VPC. The main route table is created alongside the VPC and applies to all subnets that haven't been explicitly assigned a custom route table.

Because the main route table may already have a route to the internet gateway — or could have one added in the future — which would expose the private subnet to the internet. By creating a dedicated route table for the private subnet with no internet gateway route, we explicitly control and lock down its routing behavior, independent of any changes made to the main route table.

The private subnet's route table only allows local traffic — meaning communication within the VPC's CIDR block (e.g. 10.0.0.0/16). There is no route to an internet gateway, so no inbound or outbound internet traffic is permitted. Resources in the private subnet can only talk to other resources within the same VPC.



Yönlendirme tabloları (1/3) Bilgi
Last updated 1 minute ago
Eylemler
Yönlendirme tablosu oluştur
Find route tables by attribute or tag
Name
Yönlendirme tablosu I...
Açık alt ağ ilişkilendi...
Uç ilişkilendir...
Ana
VPC
NextWork Public Route Table
rtb-0c3239287cf896950
subnet-08aa7c9decbf134...
Evet
vpc-0ba7b62ffd36d5e9b | Ne
rtb-0136bd139e0bc5382
Evet
vpc-04436c7d1bde13595
NextWork Private Route Table
rtb-0afb824b996af06bc
subnet-0080ea030806d1...
Hayır
vpc-0ba7b62ffd36d5e9b | Ne
rtb-0afb824b996af06bc / NextWork Private Route Table
Ayrıntılar
Yönlendirmeler
Alt ağ ilişkilendirmeleri
Uç ilişkilendirmeleri
Yönlendirme yayılımı
Etiketler
Yönlendirmeler (1)
İkisi de
Yönlendirmeleri düzenle
Yönlendirmeleri filtrele
Hedef
Hedef
Durum
Yayıldı
Route Origin
10.0.0.0/16
local
Aktif
Hayır
Create Route Table

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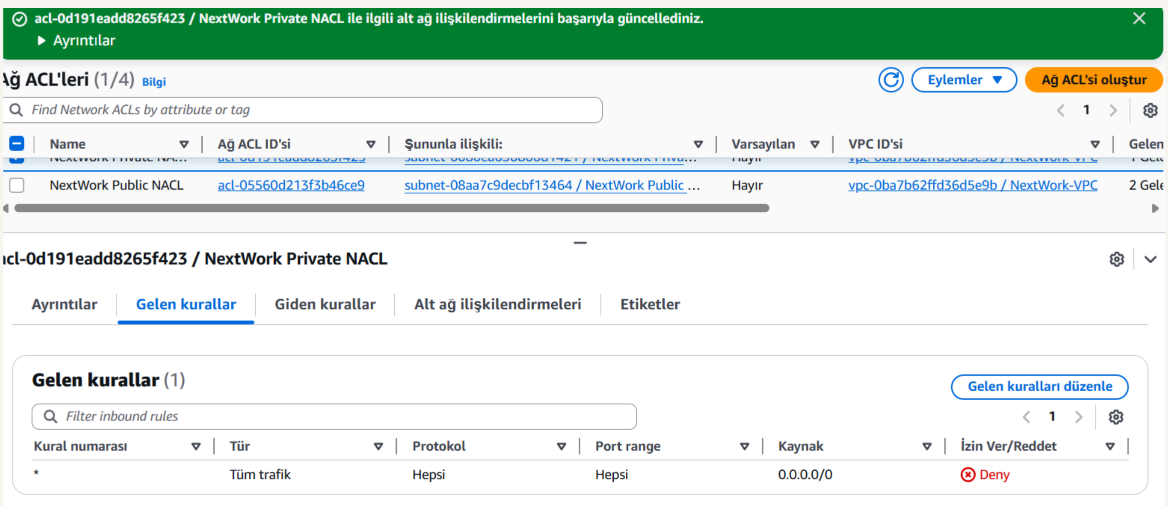
NextWork Student

A new network ACL

By default, every new subnet is automatically associated with the default NACL of the VPC. The default NACL allows all inbound and outbound traffic — which is fine for getting started, but far too permissive for a private subnet that is supposed to be isolated.

Because the default NACL allows all traffic in and out, it provides no real protection for the private subnet. By creating a custom NACL, we can explicitly define which traffic is permitted and deny everything else. This adds a dedicated layer of security at the subnet boundary, independent of security groups, following AWS's defense-in-depth approach.

Only internal VPC traffic is allowed in both directions. Everything else is denied by the catch-all * rule at the bottom. This ensures the private subnet remains completely isolated from the internet while still being able to communicate with other resources inside the VPC.



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